

Flexible Virtual Lenses to Magnify Virtual Environments Locally Without Losing Context Information

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Figure 1: Application of different lenses to the same image (a), a photograph captured at Legoland® Nagoya. From left to right: (b) a uniform zooming lens, (c) a dual zoom lens, (d) a bifocal lens and (e) another bifocal lens with a lower transition width W and a higher zooming factor Z .

ABSTRACT

We introduce virtual lenses that magnify image regions locally without losing context elements. We developed a unified formalism to design virtual lenses of any closed shape, that can be moved interactively on a screen and within a virtual reality (VR) application in real-time. Moreover, our lenses are versatile: the user can change their parameters to modify the appearance of the magnified image region dynamically. These lenses are suitable for several applications, including virtual tourism and video-surveillance. Our demo showcases an implementation of a lens changing of shape in real-time on a non-VR monitor, followed by a proof-of-concept VR version where the user can move a lens with a VR controller to deform remote textures at different distances. A video introducing both steps of our demo is available at <https://youtu.be/mMMDj67vb4>.

Index Terms: Bifocal imaging, zooming interface, virtual lens.

1 INTRODUCTION

In images containing a high density of small elements, one may magnify them by *zooming-in* to perceive their details [4]. However, this process causes contextual elements to disappear and reduces the number of simultaneous elements visible within the image (Fig. 1b).

To maintain the context, zooming interfaces introduce a movable magnified region within the image [2], acting like a *virtual lens*. As a legacy of *overview + details* and *focus + context* interfaces [5], we consider two types of virtual lenses: *dual zoom* and *bifocal* lenses. For the former, the magnified region is directly inserted into the context [3]. Then, two magnifications are displayed simultaneously (Fig. 1c). *ModulAR* inserts a circular dual zoom lens within an augmented reality (AR) environment to magnify real

world objects thanks to a modular headset with interchangeable stereo-cameras [8].

To avoid occluding scene elements, bifocal lenses add a smooth transition between the magnified region and the context [1] (Fig. 1d-e), and are especially used in image retargeting [10]. Considering a conformal mapping, bifocal lenses, computed as tridimensional (3D) meshes, can distort images while preserving local angles and minimizing distortions, at the price of a pre-computing step of several seconds [11]. According to [9], *JellyLens* is a bifocal lens that can automatically match the shape of a region of interest. However, its adaptability is limited to match filled regions or thin paths, for example, in maps.

Once a zooming interface is designed, it is often suited to magnify image elements of a specific size and shape [5], which may not be optimal for images with different contents. Therefore, we propose a highly adaptive zooming interface that increases the quantity of visible details of objects of diverse sizes and shapes. To do so, we have established a general formalism (Sec. 2), an extension of perspective walls [7] and fisheye views [6] to any closed shape. Our zooming interface consists of a freely movable bifocal lens that magnifies an image region in real time. During the first part of the demo, the user can adjust three parameters of a movable bifocal lens that changes shape across time, in parallel, at a high framerate. Then, the second part of the demo introduces a proof-of-concept application which adapts in VR a bifocal lens of a disk shape, that can be displaced on a remote texture by using the controller.

2 UNIFIED FORMALISM OF VIRTUAL LENSES

To design virtual lenses of any closed shape, we developed a general formalism and implemented it into pixel shaders. A virtual lens can modify any portion of the image that has a contiguous shape, denoted S , characterized by the radius of its circumscribed circle $R_C \in \mathbb{R}_+$. The borders of the shape can be expressed by a function $f(\theta) = R_\theta$, with $\theta \in [0, 2\pi) \subset \mathbb{N}$ and R_θ the “angular radius” of the shape. Note that if f is $C(0)$, S is a polygon and if f is at least $C(1)$, S will have a curvilinear border; comparing the two lenses of Fig. 2 illustrates that S deeply impacts the geometry of the image. Then, only the pixels inside S are projected in a different location to simulate a magnification by a dual zoom lens. To do so, the coordinates of the pixel U , $(U, V) \in \mathbb{R}^2$ are converted into the corresponding polar coordinates $U_\theta = (R, \theta)$ with $R \in \mathbb{R}_+$ and

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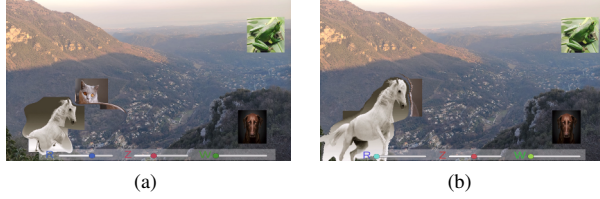


Figure 2: The first part of our demo, showcasing our non-VR zooming interface with bifocal lenses (a) of a curvilinear shape and (b) of the shape of animals.

$\theta \in [0, 2\pi[\subset \mathbb{R}$ (considering that the centroid $\mathbf{O} = (O_u, O_v)^T \in \mathbb{R}^2$ of $\mathcal{S}$ is the origin of the polar frame). Therefore, uniform scaling with a zooming factor $Z \in \mathbb{R}_+^*$, $Z > 1.0$ is applied to all pixels within the shape, that is, if $R < R_\theta$, then $\hat{R} = Z.R$ will be the radius of the warped pixel (Fig. 1c).

To simulate a bifocal lens (Fig. 1d-1e), we must add a *zooming transition*, i.e., a set of scaled versions of $\mathcal{S}$ all around the magnified shape, characterized by a *transition width* $W \in \mathbb{R}_+^*$. Within this transition, the magnification factor follows a *zooming function* $h(R) = \hat{Z}$ starting from $h(ZR) = Z$ to $h(ZR + W) = 1.0$ (without zoom), so that all elements are kept in the image, at the price of suffering geometrical distortions. Then, bifocal lenses can be parametrized by $\mathcal{P}_B = \{\mathcal{S}, h, R_C, Z, W\}$ (note that a dual zoom lens is a bifocal lens with $W = 0$).

3 SYSTEM OVERVIEW

To show the flexibility of our lenses, our demo starts with a few minutes trial of our zooming interface on a non-VR monitor, thanks to a zooming interface implemented on Unity®. On the background of Fig. 2, a curved lens (Fig. 2a) changes of shape in real time, by adjusting its local curvature at several locations at once. In addition, the lens takes the shape of four specific animals placed on the background (e.g., a horse Fig. 2b). At any time, the user can change the lens parameters by adjusting sliders on the GUI (Fig. 2).

As a proof-of-concept, the other part of our demo allows one to use a circular bifocal lens within a virtual environment. During a seated-VR experience, the user can move the lens on two different remote textures displaying a landscape (here, the background of Fig. 1). By pressing the buttons of the right controller, the user can change the radius, the zoom and the transition width of the bifocal lens. We implemented this application on Unity® and targeted it to Meta Quest 3® headsets (it can be easily adapted to other headsets). Fig. 3 shows two successive views captured from the camera of the headset, illustrating that the ray cast by the controller can interactively highlight different scene elements (e.g., a wooden boat Fig. 3a and a fake mountain Fig. 3b).

4 CONCLUSION

We introduce a unified formalism, inspired by *overview + details* and *focus + context*, to design movable virtual lenses that magnify image regions locally and that can be modified in real time. To highlight the potential of our lenses, we showcase a two-parts demo, the first one illustrating the flexibility of the lenses on a non-VR monitor, the second one proposing a proof-of-concept VR experience where the user can interactively move a bifocal lens on two remote textures. Considering this demo as a starting point, we are developing a 360 degrees version of our lenses getting closer to an immersive virtual tourism application. For this purpose, we plan to either modify the images of the rendered camera by the lenses, or the texture of a skybox, or the meshes of 3D objects.

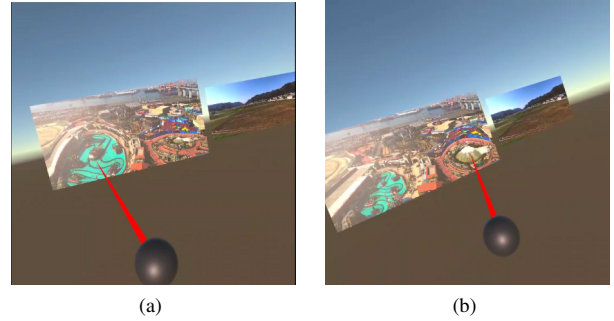


Figure 3: Two different frames captured from the headset view, while playing our virtual lens application, with two remote textures that can be deformed by the lens: (a) magnifying a boat and (b) magnifying a mountain.

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REFERENCES

- [1] C. Appert, O. Chapuis, and E. Pietriga. High-precision magnification lenses. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 273–282, 2010. 1
- [2] B. B. Bederson, J. D. Hollan, J. B. Stewart, D. Rogers, A. Druin, and D. Vick. Zooming web browser. In *Multimedia Computing and Networking 1996*, vol. 2667, pp. 260–271. SPIE, 1996. 1
- [3] H. Chang and M. F. Cohen. Panning and zooming high-resolution panoramas in virtual reality devices. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*, pp. 279–288, 2017. 1
- [4] A. D. Clark. Zoom lenses. *Monographs in Applied Optics*, 1973. 1
- [5] A. Cockburn, A. Karlson, and B. B. Bederson. A review of overview+ detail, zooming, and focus+ context interfaces. *ACM Computing Surveys (CSUR)*, 41(1):1–31, 2009. 1
- [6] G. W. Furnas. Generalized fisheye views. *Acm Sigchi Bulletin*, 17(4):16–23, 1986. 1
- [7] J. D. Mackinlay, G. G. Robertson, and S. K. Card. The perspective wall: Detail and context smoothly integrated. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pp. 173–176, 1991. 1
- [8] J. Orlosky, T. Toyama, K. Kiyokawa, and D. Sonntag. Modular: Eye-controlled vision augmentations for head mounted displays. *IEEE transactions on visualization and computer graphics*, 21(11):1259–1268, 2015. 1
- [9] C. Pindat, E. Pietriga, O. Chapuis, and C. Puech. Jellylens: content-aware adaptive lenses. In *Proceedings of the 25th annual ACM symposium on User interface software and technology*, pp. 261–270, 2012. 1
- [10] R. Wang, N. Huang, F. Tang, W. Dong, and T.-Y. Lee. Language-driven diversified image retargeting. In *SIGGRAPH Asia 2022 Posters*, pp. 1–2, 2022. 1
- [11] X. Zhao, W. Zeng, X. D. Gu, A. E. Kaufman, W. Xu, and K. Mueller. Conformal magnifier: A focus+ context technique with local shape preservation. *IEEE Transactions on Visualization and Computer Graphics*, 18(11):1928–1941, 2012. 1